

Xinyuan Yan, Ph.D.

Postdoctoral Researcher · Computational Neuroscience & Psychiatry

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EDUCATION

Postdoctoral Researcher, Neurosurgery & Neuroscience April 2025 – Present

Baylor College of Medicine *Advisors: Benjamin Y. Hayden & Sameer A. Sheth*

Postdoctoral Researcher, Psychiatry & Neuroscience March 2022 – March 2025

University of Minnesota *Advisors: Alexander Herman & David Darrow*

Ph.D., Cognitive Neuroscience September 2016 – January 2022

IDG/McGovern Institute for Brain Research, State Key Laboratory of Cognitive Neuroscience and Learning

Beijing Normal University *Advisor: Yina Ma*

B.Sc., Psychology September 2012 – June 2016

Northeast Normal University

FELLOWSHIPS & AWARDS

Travel Award, Computational and Systems Neuroscience (Cosyne), Montreal 2023

Trainee Professional Development Award, Society for Neuroscience 2022

MnDrive Neuromodulation Research Fellowship, University of Minnesota Medical School 2022

Fellowship, Computational and Cognitive Neuroscience (CCN) Summer School, CSHL Asia 2021

Fellowship, Kavli Summer Institute in Cognitive Neuroscience, UC Santa Barbara 2019

Distinguished Graduate Research Fellowship, State Key Laboratory of Cognitive Neuroscience and Learning
2018

National Graduate Research Fellowship, China 2018

PUBLICATIONS

*† indicates co-last author. * indicates co-first author.*

Concepts, language, single-neuron recordings, and hippocampus

1. **Yan, X.**, Li, J.-A., Franch, M., Zhu, H., Cowan, R., Belanger, J., Chavez, A. G., Chericoni, A., Ismail, T., Katlowitz, K. A., Kolibius, L. D., Mickiewicz, E. A., Paulo, D., Bartoli, E., Hennig, J. A., Frączek, T. M., Provenza, N. R., Rahimpour, S., Shofty, B., . . . **Hayden, B. Y.†, Sheth, S. A.†** (2026). Polysemanticity in human hippocampal neurons. *bioRxiv*. <https://doi.org/10.64898/2026.05.02.722435>
2. Zhu, H., Franch, M., Mickiewicz, E., Belanger, J., Cowan, R. L., Katlowitz, K., . . . **Hayden, B. Y.†, Sheth, S. A.†** (2026). A geometric foundation for word meaning in the brain. *bioRxiv*. <https://doi.org/10.64898/2026.01.28.702241>
3. Franch, M., Katlowitz, K. A., Mickiewicz, E. A., Belanger, J. L., Mathura, R. K., Zhu, H., . . . **Hayden, B. Y.†, Sheth, S. A.†** (2026). Neural signatures of impaired semantic contextualization in autism spectrum disorder. *bioRxiv*. <https://doi.org/10.1101/2026.03.16.712048>

4. Chericoni, A., Diao, C., **Yan, X.**, Ismail, T., Mickiewicz, E. A., Franch, M., . . . **Hayden, B. Y.†, Sheth, S. A.†** (2026). Neural geometry in the human hippocampus enables generalization across spatial position and gaze. *arXiv*. <https://arxiv.org/abs/2603.04747>
5. **Yan, X.**, Chavez, A. G., Franch, M., Katlowitz, K. A., Gautam, I., Kim, B., Krishna, A., Shrivastava, A., Van Arsdell, K., Belanger, J., Chericoni, A., Ismail, T., Mickiewicz, E. A., Paulo, D., Zhu, H., Goldman, A. M., Krishnan, V., Maheshwari, A., Bartoli, E., . . . **Hayden, B. Y.†, Sheth, S. A.†** (2025). Shared neural geometries for bilingual semantic representations in human hippocampal neurons. *bioRxiv*. <https://doi.org/10.1101/2025.11.16.688726> [Accepted in principle, *Cell*]
6. Mickiewicz, E. A., Franch, M., Katlowitz, K. A., Chavez, A. G., Zhu, H., Chericoni, A., . . . Hayden, B. Y. (2025). A semantotopic map in the human hippocampus. *bioRxiv*. <https://doi.org/10.1101/2025.10.31.685959>
7. Chavez, A. G., Franch, M., Mickiewicz, E. A., Baltazar, W., Belanger, J. L., Devara, D., . . . Hayden, B. Y. (2025). Mirror manifolds: Partially overlapping neural subspaces for speaking and listening. *bioRxiv*. <https://doi.org/10.1101/2025.09.20.677504>

Beliefs about uncertainty (explore–exploit) and implications for affective disorders

Skill set: computational modeling, intracranial EEG

8. **Yan, X.**, Peterson, C. M., Schmidt, L. M., Koenig, S., Nixdorf, D. R., Herman, A., & Darrow, D. (2025). Impaired adaptive learning in chronic pain contributes to apathy. *eLife*.
9. **Yan, X.**, Ebitz, B. R., Darrow, D. P., & Herman, A. B. (2025). Nonlinear modulation of human exploration by distinct sources of uncertainty. *bioRxiv*, 2025-04. [Under review at *PNAS*]
10. **Yan, X.**, König, S. D., Ebitz, R. B., Hayden, B. Y., Darrow, D. P., & Herman, A. B. (2025). Dynamic prefrontal coupling coordinates adaptive decision-making. *Research Square*, rs-3. [Under review at *Nature Communications*]
11. **Yan, X.**, Ebitz, R. B., Grissom, N., Darrow, D. P., & Herman, A. B. (2025). Distinct computational mechanisms of uncertainty processing explain opposing exploratory behaviors in anxiety and apathy. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*.
12. Knap, E., **Yan, X.**, Chen, C. S., Jacob, S., Darrow, D. P., Ebitz, R. B., . . . Herman, A. B. (2025). Social aloofness is associated with non-social explore–exploit decisions. *Communications Psychology*, 3(1), 106.
13. Abbaszadeh, M., Ozanick, E., Magen, N., Darrow, D., **Yan, X.**, Grissom, N., . . . Ebitz, B. R. (2025). Individual differences in sequential decision-making. *bioRxiv*. [Under review at *iScience*]
14. Cao, D.*, Cui, J.*, **Yan, X.***, Zhang, X., Dang, Y., Zhao, H., . . . Zhang, Y. (2025). Dynamic interactions in human amygdala subregions support hierarchical processing of fearful faces. *bioRxiv*, 2025-11.

Social decision-making and its modulation

Skill set: computational modeling, fMRI

15. Kappes, A., **Yan, X.**, Crockett, M. J., & Ma, Y. (2025). Cultural differences in vicarious optimism. *Personality and Social Psychology Bulletin*, 01461672251319566.
16. **Yan, X.** (2024). Too smart for one's own good? Evolution favors individuals with a low-level theory of mind in a complex network. *PsyArXiv*.
17. Zheng, S.†, **Yan, X.†**, Siegel, J., Chituc, V., Li, S., Crockett, M., & Ma, Y. (2020). Self-serving karmic beliefs: Prosociality influences vicarious optimism. *PsyArXiv*. [† co-first author]
18. Liu, Y., Li, S.#, Lin, W.#, Li, W.#, **Yan, X.**, Wang, X., Pan, X., Rutledge, R. B., & Ma, Y. (2019). Amygdala activity tracks deviations from social preferences. *Nature Neuroscience*, 22(4), 633–641. [# co-second author]
19. **Yan, X.**, Yong, X., Huang, W., & Ma, Y. (2018). Placebo treatment facilitates social trust and approach behavior. *PNAS*, 115(22), 5732–5737.

20. Wang, D., **Yan, X.**, Li, M., & Ma, Y. (2017). Neural substrates underlying oxytocin effect: A quantitative meta-analysis of pharmaco-imaging studies. *Social Cognitive and Affective Neuroscience*, 12(10), 1565–1573.

Self-representation and implications for major depressive disorder

Skill set: fMRI

21. **Yan, X.** (2024). Neural dynamics of self-referential processing and the insight for decoding self-concepts. *Journal of Neuroscience*, 44(30). [Commentary]
22. Feng, C.†, **Yan, X.†**, Huang, W., Han, S., & Ma, Y. (2018). Neural representations of the multidimensional self in the cortical midline structures. *NeuroImage*, 183, 291–299. [† co-first author]
23. Ma, Y., Li, S., Wang, C., Liu, Y., Li, W., **Yan, X.**, Chen, Q., & Han, S. (2016). Distinct oxytocin effects on belief updating in response to desirable and undesirable feedback. *PNAS*, 113(33), 9256–9261.

Cortisol and the stress model

Skill set: resting-state fMRI

24. Shang, G., Zhou, T., **Yan, X.**, He, K., Liu, B., Feng, Z., . . . Zhang, Y. (2025). Multiscale analysis reveals hippocampal subfield vulnerabilities to chronic cortisol overexposure: Evidence from Cushing's disease. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*.
25. Feng, Z., Zhou, T., **Yan, X.**, He, K., Liu, H., Yu, X., Lu, R., Ma, Z., Yu, X., & Zhang, Y. (2025). Selective hippocampal subfield atrophy mediates cognitive decline in Cushing's disease. *Brain and Behavior*, 15(11), e71030.
26. Shang, G., Zhou, T., Yu, X., **Yan, X.**, He, K., Liu, B., Feng, Z., Zhang, Y., & Yu, X. (2024). Chronic hypercortisolism disrupts the principal functional gradient in Cushing's disease: A multi-scale connectomics and transcriptomics study. *NeuroImage: Clinical*, 43, 103652.
27. Liu, B., Mao, Z., **Yan, X.**, Yang, H., Xu, J., Feng, Z., Zhang, Y., & Yu, X. (2024). Structural network topologies are associated with deep brain stimulation outcomes in Meige syndrome. *Neurotherapeutics*, 21(4), e00367.
28. Feng, Z., Zhang, H., Zhou, T., Yu, X., Zhang, Y., & **Yan, X.*** (2023). Dynamic functional connectivity changes associated with psychiatric traits and cognitive deficits in Cushing's disease. *Translational Psychiatry*, 13(1), 308. [* corresponding author]

EDITORIAL SERVICE

- **Guest Associate Editor, *Frontiers in Human Neuroscience*** — Research Topic: *Methods in Cognitive Neuroscience: 2023* (Oct 2023 – Dec 2023)

AD-HOC REVIEWER

PNAS · Nature Communications · Communications Biology · Translational Psychiatry · Brain and Behavior · Psychoradiology (Oxford Academic) · Self and Identity · Frontiers in Psychology · Frontiers in Psychiatry · Frontiers in Human Neuroscience · Frontiers in Endocrinology · Frontiers in Robotics and AI · BMC Neuroscience.

PROFESSIONAL MEMBERSHIPS

- Society for Neuroscience
- Society of Biological Psychiatry (Regular Member)
- Sigma Xi

CONFERENCE PRESENTATIONS

Oral Presentations

- **Yan X**, Koneig S, Ebitz B, Hayden, B, Darrow D, Herman A. *Dynamic prefrontal coupling coordinates adaptive decision-making*, Society for Neuroscience, November 15-19, 2025, San Diego, USA.
- **Yan, X.**, König, S., Ebitz, B., Hayden, B., Darrow, D., & Herman, A. (2024). *Deciphering neural mechanisms of explore–exploit decision-making in the prefrontal cortex*. Computational Psychiatry Conference, Minneapolis, USA.
- **Yan, X.**, Ebitz, B., . . . Darrow, D., & Herman, A. (2022). *The behavioral profile of the exploration–exploitation dilemma and its implications for anhedonia and anxiety*. Computational Psychiatry, New York City, USA.
- **Yan, X.** (2020). *Neural representations of social placebo effect*. Neuromatch 3.0. [Recording](#).

Poster Presentations

- **Yan, X.**, König, S., Ebitz, B., Darrow, D., & Herman, A. (2023). *Intracranial electrophysiological evidence for a novel neuro-computational mechanism of exploration–exploitation decision-making in humans*. Computational and Systems Neuroscience (Cosyne), Montreal, Canada.
- **Yan, X.**, König, S., Ebitz, B., Darrow, D., & Herman, A. (2022). *Intracranial electrophysiological evidence for a novel neuro-computational mechanism of exploration–exploitation decision-making in humans*. Society for Neuroscience Annual Meeting, San Diego, USA.
- **Yan, X.**, & Ma, Y. (2018). *Neural pattern change of memory control of personal strengths and weaknesses*. 11th Annual Meeting of the Social & Affective Neuroscience Society, New York City, USA.
- Ma Y, **Yan X**, Huang W. *Placebo effects on human social trust and approach behavior*, 2018, 11th Annual Meeting of The Social & Affective Neuroscience Society. New York City, USA.

TECHNICAL SKILLS

- **Programming:** MATLAB (advanced), R, Python, HTML5/CSS3 (basic)
- **Methods:** Single-unit intracranial recordings, Neuropixels, intracranial EEG, fMRI (task & resting-state), spike sorting, computational modeling, representational similarity analysis (RSA), reinforcement learning, neural population geometry
- **Visualization:** Adobe Illustrator (scientific figures, with formal training)

ADDITIONAL TRAINING

- Computational and Cognitive Neuroscience (CCN) Summer School, *Cold Spring Harbor Laboratory*, 2021
- Online Summer School for Computational Neuroscience, *Neuromatch Academy*, 2020
- Kavli Summer Institute in Cognitive Neuroscience, *UC Santa Barbara*, 2019
- Coursera coursework (2020): Computer Simulations · Social Network Analysis · Network Dynamics of Social Behavior · Julia Scientific Programming · Facial Expression Recognition with Keras · Introduction to HTML5

TEACHING

Mentoring junior graduate students in core skills for scientific research:

- Representational similarity analysis
- Resting-state fMRI analysis
- Diffusion decision model & value-based DDM
- Reinforcement learning & Markov decision processes

- Attractor networks & decision-making
- Neural networks — basic concepts
- Building a personal academic website (HTML5 + CSS3)